D 100636 **(Pages : 2)** **Name.....**

Reg. No.....

SIXTH SEMESTER U.G. (CBCSS–UG) DEGREE EXAMINATION, MARCH 2024

Physics/Applied Physics

PHY6B12/APH 6B 12—NUCLEAR PHYSICS AND PARTICLE PHYSICS

(2019 Admission onwards)

Time : Two Hours Maximum : 60 Marks

The symbols used in this question paper have their usual meanings.

Section A – Short Answer type

Answer **all** *questions in two or three sentences, each correct answer carries a maximum of* 2 *marks.*

1. Explain the correction in the binding energy formula obtained using liquid drop model.
2. Why do heavy nuclei have more neutrons than protons ?
3. Write a short note on radio isotope production in nuclear reaction.
4. What is the strange behavior of kaons and hyperons ?
5. What is the working principle of an intersecting beam accelerator ?
6. Give the list of leptons. Mention the charge of each particle.
7. What do you mean by quantum chromodynamics ?
8. In general, would you expect fission fragment to decay by positive or negative beta decay ? Why ?
9. Comment on the property of nuclear force.
10. List some similarities and difference between the properties of photons and neutrinos.
11. Explain the working of semiconductor counters.
12. Explain why a fusion reactor requires a high particle density, a high temperature and a long confinement time ?

(Ceiling 20 marks)

Section B - Paragraph / Problem type

Answer **all** *questions in a paragraph of about half a page to one page, each correct answer carries a maximum of* 5 *marks.*

13. Consider a single helium nucleus formed by the fusion of two deuterium nuclei. Mass of ${}_1H^2 = 2.014102u$; mass of ${}_2He^4 = 4.002604u$. Find out the energy released in fusion.
14. Distinguish between fission and fusion reactions. Explain the fusion process in stars.

Turn over

15. Determine whether the following reactions are allowed or forbidden.

    (i) $p + p \rightarrow n^{+} + p + \pi^{+}$.

    (ii) $p + \pi^{-} \rightarrow \pi^{0} + n$.

    (iii) $e^{+}e^{+} \rightarrow \mu^{+} + \pi^{-}$.

16. Discuss briefly low energy reaction kinematics.

17. The disintegration constant λ of a radioactive element is 0.00231 per day. Calculate its half-life and average life.

18. Discuss the Quark model.

19. A reactor is developing energy at the rate of 3000kW. How many atoms of U^{235} undergo fission per second ?

(Ceiling 30 marks)

Section C - Essay type

Essays - Answer in about two pages, any **one** *question.*
Answer carries 10 *marks.*

20. Using a neat diagram explain the working principle of Van de Graaff electrostatic generator.

21. Explain the different elementary particle quantum numbers and their conservation laws with examples.

(1 × 10 = 10 marks)